

FAKE PROFILE IDENTIFICATION IN SOCIAL NETWORK USING MACHINE LEARNING AND NLP

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Abstract— Now-a-days people are using social media by creating their unique profiles for Communication with Friends & Family. But there are some threats to the user's Privacy and Security. To know who is creating those threats, we employ machine learning and natural language processing techniques to categorize each user's unique profile. Here we identify the fake profile using Machine Learning models. This project proposes a classification approach to distinguish genuine and fake profiles, addressing security challenges in the prevalent realm of social network platforms.

Keywords: classification, natural language processing, and machine learning.

I. INTRODUCTION

Fake profiles on social networks, which can cause harm like identity theft and spreading false information. It aims to build a smart computer system using machine learning and language processing to spot these fake profiles accurately. Unlike previous methods, it will use advanced algorithms like Support Vector Machines, Naive Bayes and KNN to improve accuracy. By studying a wide range of social profiles, the system will learn to recognize patterns that indicate fakeness. This project isn't just academic; it aims to make the internet safer by helping social networks better identify and remove fake accounts. By the end, it hopes to create a system even better than the current one and share what it learns to make social networks safer for everyone.

II. LITERATURE REVIEW

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- These sources provide insights into the methods and techniques used for detecting faked web profiles leveraging machine learning and natural language processing. It is evident that researchers are actively exploring ways to combat the problem with bogus profiles and enhance the genuineness of social media platforms.

III. EXISTING SYSTEM

It is probable that the current model for detecting deceptive profiles uses a Random Forest technique. An algorithm for supervised learning is called random forest. An assembly of decision trees, often trained using the bagging approach, makes up the "forest" it creates. The widely held belief behind the bagging technique is that combining various skillfully executed looks would improve the overall outcome. Although the current model might use some basic machine learning algorithms, it might not make use of more powerful natural language processing or deep learning approaches.

IV. DISADVANTAGES OF EXISTING SYSTEM

- Interpretability: Compared to a single decision tree, Random Forests can be more difficult to understand because they are an ensemble of several decision trees.
- Computational Complexity: Training a huge Random Forest with several decision trees can be computationally expensive and time-consuming.
- Handling Noisy Data: Struggles with outliers or noisy data.

V. PROPOSED SYSTEM

- The proposed model introduces machine learning and natural language processing concepts like SVM, KNN, Naive Bayes, Lemmatization, Stop Word, Stemming.
- The aim is to enhance the accuracy rate at which social network phony profiles can be identified.
- The proposed model seeks to address the limitations of Random Forest methods and improve the overall effectiveness of fake profile detection.

VI. PROPOSED SYSTEM ARCHITECTURE

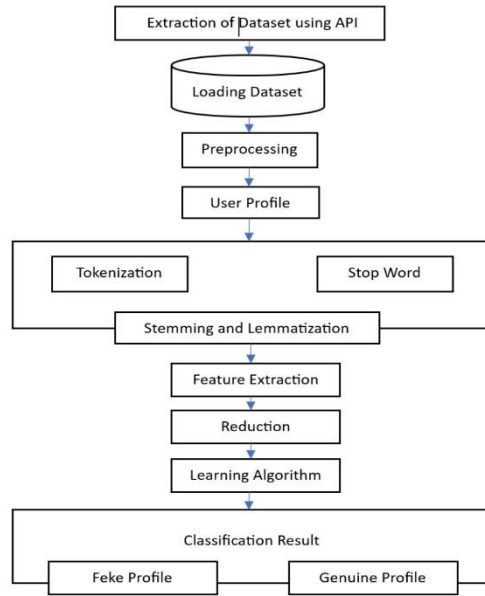


Fig. 1 Proposed System Architecture

VII. PROPOSED SYSTEM METHODOLOGY

1. *Data Collection and Preprocessing:* Identify sources of training data, including text and image datasets. Some common sources are like public datasets, web scraping, government databases and many more. Implement data collection processes to gather diverse and representative samples. Gather information from a wide range of people and places to make sure it represents everyone fairly and accurately. Use different methods like surveys or interviews to collect this information in a respectful and careful way. Apply preprocessing steps, such as text tokenization, feature extraction. The necessary libraries are imported at the beginning, including pandas for data manipulation, Streamlit for building the web application, scikit-learn for machine learning tasks, NLTK for natural language processing, and joblib for saving models. Dataset is loaded from a CSV file. Missing values in certain columns are filled with empty strings. Features like name, follower count, friend count, location, and description are combined into a single feature called combined_features. Text preprocessing is performed on the combined features. This includes tokenization, removing stopwords, punctuation, and applying lemmatization.
2. *Model Training and Testing:* Training several classifiers, each geared towards identifying fake profiles on social networks. These classifiers include Support Vector Machines (SVM), Naive Bayes, Random Forest, and K-Nearest Neighbors (KNN). Before training, the data undergoes preprocessing to clean it up and reduce its complexity. This preprocessing step helps in improving the performance of the classifiers. Additionally, we reduce the dimensionality of the data, making it easier for the classifiers to work with. By training these classifiers on the pre-processed and dimensionality-reduced data, we aim to enhance their accuracy in detecting fake profiles. Each classifier learns from the patterns present in the data, such as user behaviour and linguistic cues, to distinguish between genuine and fraudulent accounts. This approach allows us to explore different techniques and algorithms to find the most effective solution for identifying fake profiles.
3. *Model Evaluation and Persistence:* Once the classifiers have predicted labels for the testing set, we calculate and print out evaluation metrics in order to evaluate the performance of each exemplar. Depending on the particular requirements of the project, these measures may also include recall, accuracy, precision, and F1-score. These metrics provide insights into the

degree to which each model is functioning in term of recognizing deceptive profiles. After evaluating the models, we save the trained models, including the vectorizer, SVM, Naive Bayes, KNN, and PCA (Principal Component Analysis), using joblib. This allows us to use these models in the future without needing to retrain them. Saving the models ensures that we can deploy them in real-world scenarios efficiently and consistently. By saving the trained models, we ensure that the efforts put into training and fine-tuning them are preserved, enabling us to apply them to new data and tasks effectively. This approach streamlines the process of model deployment and facilitates the scalability of the solution to larger datasets or different social networks.

4. *Utilizing the Top-Performing Algorithm for Predicting Fake or Genuine Profiles on Social Networks:* Evaluating the performance of the classifiers and determined which algorithm achieved the highest accuracy, we can proceed to use that algorithm for predicting whether a profile is fake or genuine. This algorithm, which demonstrates the best performance in terms of accuracy, will be the most reliable choice for making predictions on new or unseen data. Using the selected algorithm, we can deploy a prediction system that takes as input the features of a profile and outputs a prediction of its authenticity (fake or genuine). This prediction system can be integrated into social network platforms to automatically flag suspicious profiles or assist users in making informed decisions about the profiles they interact with. By leveraging the algorithm with the highest accuracy, we can ensure that our prediction system provides the most reliable and accurate assessments of profile authenticity, thereby upgrading the overall assurance and reliability of social networks for users.

VIII. ADVANTAGES OF PROPOSED MODEL

The proposed strategy to recognize fraudulent profiles in social networks making use of machine learning and natural language processing (NLP) offers several advantages through the use of three classification techniques: Support Vector Machine, Naïve Bayes, and K-Nearest Neighbors.

1. *Enhanced Accuracy:* Utilizing sophisticated algorithms such as Naive Bayes, KNN, and Support Vector Machines, the system can detect fraudulent profiles with greater precision than traditional approaches. The result makes it possible to identify and delete bogus accounts from social networks with greater accuracy.
2. *Robustness:* The system's ability to study a wide range of social profiles allows it to learn diverse patterns indicative of fake accounts. This increases its robustness in detecting various types of fake profiles, including those employing sophisticated techniques to appear genuine.
3. *Real-time Detection:* With the combination of language processing methods and machine learning, the system can perform detection of phony profiles in real time as they are created or engage in suspicious activities on social networks. This proactive approach helps mitigate potential risks to users' privacy and security.
4. *Scalability:* The proposed system, based on machine learning algorithms, is scalable and adaptable to evolving threats and trends in fake profile creation. It can handle large volumes of data and profiles, making it suitable for deployment on popular social network platforms with millions of users.
5. *Contribution to Internet Safety:* By assisting social networks in better identifying and removing fake accounts, the proposed system contributes to making the internet safer for users. It helps combat issues such as identity theft, misinformation, and illicit activities propagated through fake profiles, thereby fostering a more trustworthy online environment.
6. *Knowledge Sharing:* The project's aim to share insights and findings with social network platforms promotes collaboration and collective efforts in addressing the challenge of fake profiles. By disseminating learnings, the proposed system can catalyze improvements in existing detection mechanisms across various online platforms.

IX. RESULT

Accuracy Rating:

Accuracies:

SVM Classifier Accuracy: 0.9343971631205674

Naive Bayes Classifier Accuracy: 0.9148936170212766

KNN Classifier Accuracy: 0.8847517730496454

Fig. 2 Accuracy Rating

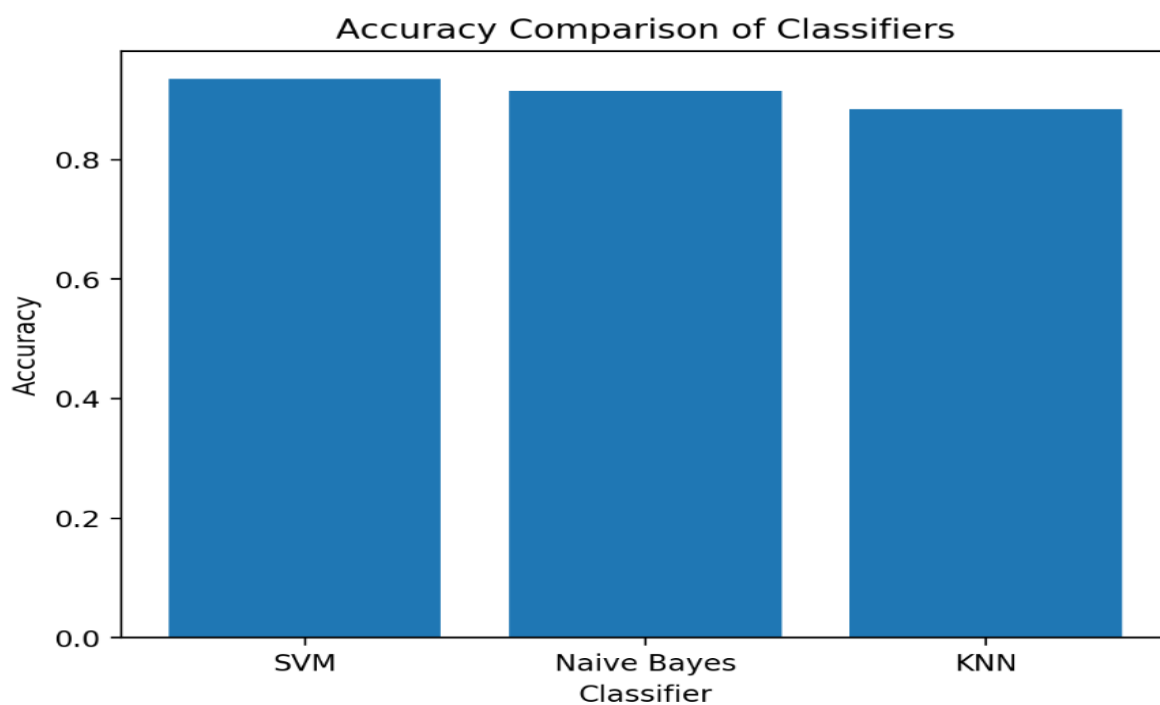


Fig. 2 Accuracy Rating Plot

Confusion matrices:

Confusion Matrices:

SVM Classifier:

	0	1
0	255	1
1	35	273

Naive Bayes Classifier:

	0	1
0	233	23
1	29	279

KNN Classifier:

	0	1
0	252	4
1	62	246

Fig. 3 Confusion matrices

X. CONCLUSION

This project focuses on identifying fictitious online profiles by using advanced computer techniques like machine learning and natural language processing (NLP). We trained algorithms like Support Vector Machines (SVM), Naive Bayes, and K-Nearest Neighbors (KNN) on a dataset containing real and fake profiles' features. The algorithms learned to identify patterns that distinguish fake from genuine profiles.

Our project significantly improved the ability to detect fake profiles on online forums. This is essential for ensuring trust and honesty in online communities, as it helps prevent deceptive practices and maintains a safe and authentic environment for users.

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